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V300 ZEFIRO

A New Sense of Speed

Fire Protection and Extinction-VF.CONCEPT SFRS

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Revision	Date	Changes	Reason
00	16.09.2011		
01 Generated from DOORS module /2F_06.04.-.Fire Protection and Extinction-VF.-.Zefiro-Europe/Fire Protection and Extinction-VF.CONCEPT, baseline 1.2	20.04.2012	1. Modified 2F_06.04.Zefiro-Europe.CONCEPT.363. 2. Modified 2F_06.04.Zefiro-Europe.CONCEPT.175 and added signal _Fire_Onboard_Received_DM8 at section 4.1.1 3. Modified 2F_06.04.Zefiro-Europe.CONCEPT.80 as concerns “Derived to” and “User interface” 4. pictures at section 3.3.1.1 corrected 5. TCMS signals required by 2F_06.04.Zefiro-Europe.CONCEPT.190 and 2F_06.04.Zefiro-Europe.CONCEPT.206 added at section 4.2 6. Removed 2F_06.04.Zefiro-Europe.CONCEPT.471 and modified 2F_06.04.Zefiro-Europe.CONCEPT.293 7. Modified 2F_06.04.Zefiro-Europe.CONCEPT.83 and 2F_06.04.Zefiro-Europe.CONCEPT.221 8. 2F_06.04.Zefiro-Europe.CONCEPT.182 deleted and 2F_06.04.Zefiro-Europe.CONCEPT.183, 2F_06.04.Zefiro-Europe.CONCEPT.84, 2F_06.04.Zefiro-Europe.CONCEPT.223 modified	1. Event indication missing 2. Signal _Fire_Onboard_Received_DM8 missing 3. The mute button for Alarm is moved to TCMS IDU after request from TI 4. Ethernet line for the slave control unit missing; NC contact in picture 4 not correct 5. signals missing at section 4.2 6. the status of fire general alarm is unique at train level 7. Pre-activation status coincides with Alarm status (pass. areas) and Discharge disabled status added 8. Pre-alarm status is related to a coach and not to a single detector; sensor disabled status added

02 Generated from DOORS module /2F_06.04.-.Fire Protection and Extinction-VF.-.Zefiro-Europe/Fire Protection and Extinction-VF.CONCEPT, baseline 1.3	30.11.2012	Modified 2F_06.04.Zefiro-Europe.CONCEPT.198, 199, 200, 201, 202 and 2F_06.04.Zefiro-Europe.CONCEPT.270	Names of the gangway doors have been specified at half train level instead of at train level
03 Generated from DOORS module /Platform.-.Zefiro-Europe/01.-.Requirements Breakdown Structure.-.Zefiro-Europe/2F.-.Vehicle-Functional Integration Areas.-.Zefiro-Europe/2F_06.-.Auxiliaries-VF.-.Zefiro-Europe/2F_06.04.-.Fire Protection and Extinction-VF.-.Zefiro-Europe/Fire Protection and Extinction-VF.CONCEPT, current 1.3	10.06.2015	Modified 2F_06.04.Zefiro-Europe.CONCEPT.83, 221, 242, 84, 223, 224, 363. Removed 2F_06.04.Zefiro-Europe.CONCEPT.475, 234, 148, 149. Added 2F_06.04.Zefiro-Europe.CONCEPT.479, 480, 481, 482, 483, 484, 485, 487.	Document has been aligned to ICD 3EST000225-7440 rev. E. Out of order icon functionality has been specified. Events related to variable IGenSystemMode have been specified. Vehicle interfaces commands in case of fire in vestibules/toilets have been aligned to System Functional Description.

04	02/12/2020	§2.1: description updated §3.3.4.2: modified following concepts: 2F_06.04.Zefiro-Europe.CONCEPT.83, 2F_06.04.Zefiro-Europe.CONCEPT.221 §3.3.5 – 3.3.5.6: complete section marked as not applicable for ETR1000 Spain §3.4: description updated §3.4.2: Modified following concepts: 2F_06.04.Zefiro-Europe.CONCEPT.233 2F_06.04.Zefiro-Europe.CONCEPT.236 2F_06.04.Zefiro-Europe.CONCEPT.237 2F_06.04.Zefiro-Europe.CONCEPT.88 2F_06.04.Zefiro-Europe.CONCEPT.238 2F_06.04.Zefiro-Europe.CONCEPT.239 2F_06.04.Zefiro-Europe.CONCEPT.240 2F_06.04.Zefiro-Europe.CONCEPT.241 2F_06.04.Zefiro-Europe.CONCEPT.242	Document has been updated to be applicable also to ETR1000 Spain project (water mist system not present and HVAC protection devices (LHC and aerosol) not installed).

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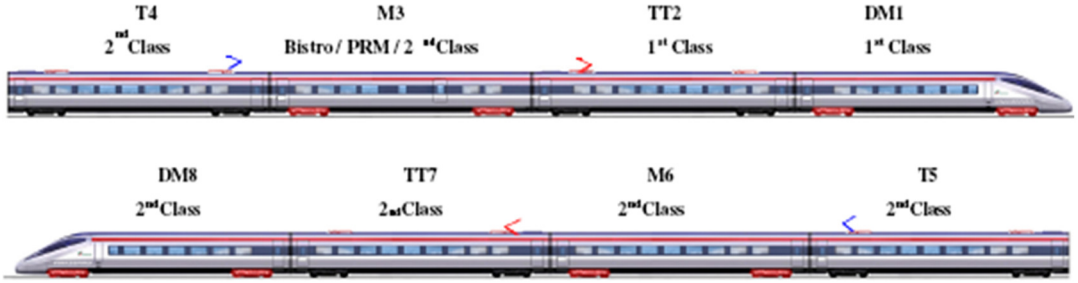
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1 Introduction

This document is a function specification for the control of the Fire Detection and Extinguishing system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
FPS	Fire Protection System
HW HMI	Hardware Human Machine Interface, e.g. buttons, swiches and indication lamps.
IDU	Interface Display Unit
LHC	Linear Heat Cable
MAU	Machinery Accumulator Unit
NO	Normally Open
PB	Push Button
PS	Power Supply
TCMS	Train Control and Management System

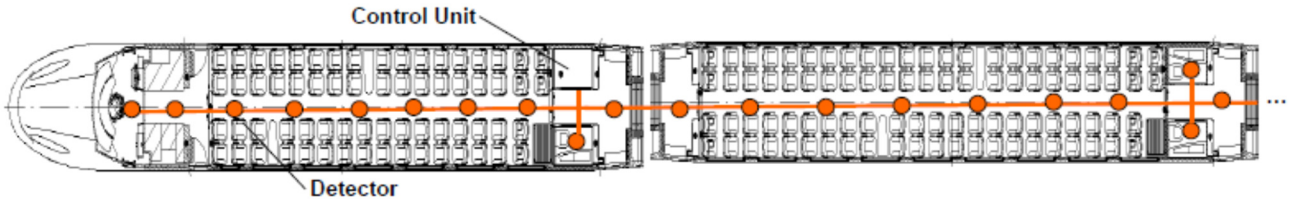
2 System Overview

2.1 System Configuration

The Fire Protection System (FPS) is the result of integrating 3 different parts of the system:

- fire detection;
- water mist fire suppression in passenger areas (not applicable to ETR1000 Spain);
- aerosol extinguishing in technical areas.

Fire detection is performed everywhere in passenger areas (including vestibules, toilets and bistrò) and staff areas (including staff toilet, driver cabs and electronic rooms). In these zones the fire is detected by using point detectors capable of detecting the presence of smoke and the temperature increase.
In the Picture 1 below the position of such point detectors and the control unit is shown in the first 2 coaches; in the other coaches the devices will be positioned symmetrically except for DM3 which will present a special configuration of point detectors due to the particular layout of the coach.



Picture 1 - Point detectors and control unit layout

Moreover fire detection is performed in all HV and MV cubicles installed underfloor and on the roof. Here below is the list of such protected areas; in ETR1000 Spain the E-cubicles inside the HVAC (position 13 of the table) will not protected with fire detection and extinguishing system:

No.	Position	Name	Total number of boxes on 8-car train	Locations	Must be detected	Must be extinguished
1	Underfloor	AC Control Box (Transformer Control Box)	2	TT2, TT7	Yes	Yes
2	Underfloor	DC Main Circuit Breaker	2	TT2, TT7	Yes	Yes
3	Underfloor	Line Converter (LL)	4	DM1, M3, M6, DM8	Yes	Yes
4	Underfloor	Converter Cooler (Cooling)	4	DM1, M3, M6, DM8	Yes	Yes
5	Underfloor	Motor Converter (4M)	4	DM1, M3, M6, DM8	Yes	Yes
6	Underfloor	Medium Frequency Module (FF)	4	DM1, M3, M6, DM8	Yes	Yes
7	Underfloor	Auxiliary Converter (AE)	4	DM1, M3, M6, DM8	Yes	Yes
8	Underfloor	Battery Charger	4	T4, T5	Yes	Yes
9	Underfloor	(AC) Distribution Box	8	in every car	Yes	Yes
10	Underfloor	Transformer Charging Box (Transformer Side Box)	2	TT2, TT7	Yes	Yes
11	Underfloor	Transformer & Tranformer Cooler	2	TT2, TT7	Yes	no
12	Roof	HV Box	2	T4, T5	Yes	Yes
13	Roof	E-Cubicle inside HVAC	18	Twice in every car, 1 more on each cab	Yes	Yes

In these zones the fire is detected by using linear heat cables (LHC) that are heat sensitive and in case the set temperature is reached give an electrical signal to the control unit.
The water mist fire suppression system is performed in all passenger and staff areas except for the driver cabs and electronic rooms where no sprinklers are installed and a fire will only activate an alarm to TCMS.
The suppression system is designed at train level: the water/gas cylinders (MAU units) will be installed only in first and last coaches and there will be a pipe network with a main branch, going along the whole train and connecting the 2 MAU units, and suitable derivations to the sprinklers. The sprinklers contain a heat sensitive bulb that breaks in case of high temperature and discharge the water mist only locally.
The suppression system is commanded by the detection system that activates both MAU units in case of fire.

The extinguishing system in technical areas is performed by using aerosol cartridges installed in each enclosure listed in the table. The extinguishing system is commanded by the detection system that activates the aerosol of the enclosure where fire has been detected (after a pre definite delay in order to let possible fan to stop).
The fire protection system is connected to the driver desks via HW signals in order to light a lamp in case of a fire general alarm and to TCMS via Ethernet in order to exchange status and diagnostic information.
The fire protection system is connected to HVAC systems via HW signals in order to shut down ventilation and close external shutters of the HVAC units of the coach where the fire has been detected.
Moreover it is connected to gangway doors via HW signals in order to close both gangway doors of both gangways of the coach where the fire has been detected.
Lastly it is connected to suitable electric components in order to disconnect power in case of fire.
In case of coupled trains two hardwired signals are foreseen over the coupler in order to transmit the status of the fire general alarm from one train to the other and allow the lighting of the lamp also on the coupled train driver desks.

3 Functional requirements

3.1 System configuration

3.1.1 Function Design

Fire detection is performed in passenger saloons, vestibules, toilets, bistrò and staff areas (including staff toilet, driver cabs and electronic rooms). In these zones the fire is detected by using point detectors capable of detecting the presence of smoke and the temperature increase.

Moreover fire detection is performed in all HV and MV cubicles installed underfloor and on the roof (see table at section 2.1). In these zones the fire is detected by using linear heat cables (LHC).

3.1.1.1 Normal Condition

The detectors installed in passenger saloons, vestibules, toilets, bistrò and staff areas provide continuously to the control units a analogical measure of the smoke and temperature level of the supervised area. In case the level of smoke and/or temperature is higher than a pre definite threshold the control unit gives a Pre-alarm condition followed by the Alarm condition, if reached.
Each passenger area is covered by more than one detector except for the toilets where only one detector is installed.
The detectors installed in the cubicles under the vehicle floor or on the roof provide continuously to the control units a digital information regarding their status. If the temperature is higher than the set level, the detector closes a contact and the control unit gives the Alarm status. Each LHC provides and end of line resistor.

In each cubicle only one detector is installed.

3.1.1.2 Failure Cases

a) Loss of function of a smoke/temperature detector in passenger/staff areas.

b) Loss of function of a temperature detector (LHC) in technical areas.

3.1.1.3 Failure Detection

a) Failure condition transmitted to TCMS.

b) Failure condition transmitted to TCMS.

3.1.1.4 Failure Handling

Considering that smoke or temperature part of the detector will probably continue to work and that in the passenger area the function is realized also by the adjacent detectors, a redundancy is lost but the

function is not endangered. As far as a failure of the detector in toilet is concerned the function is endangered and the driver shall act according to the operational manual.

Temperature detectors in technical areas are not redundant, therefore the function is endangered and the driver shall act according to the operational manual.

3.1.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.96
Description	Smoke/fire shall be detected in toilets.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.67	One smoke/temperature detector shall be foreseen in each toilet.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.133	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.97
Description	Smoke/fire shall be detected in the driver cabs.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.68	One smoke/temperature detector shall be foreseen in both driver cabs.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.134	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.98
Description	Smoke/fire shall be detected in passenger saloons, vestibules and gangways.
Safety level	SIL 2

No smoke/temperature detectors will be installed in gangways. Due to the fact that gangway doors are normally open, fire detection in gangways will be performed by smoke/temperature detectors installed in vestibules.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.69	A suitable number of smoke/temperature detectors shall be foreseen in passenger saloons and vestibules.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.135	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.101
Description	Smoke/fire shall be detected in technical areas where MV or HV devices are installed.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL

2F_06.04.Zefiro-Europe.CONCEPT.70	Suitable temperature detectors shall be foreseen to detect possible fire arising in technical boxes containing MV and HV components.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.136	All thermal detectors for technical areas shall communicate to the control units their status.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.106
Description	Smoke/fire shall be detected in electronic room.
Safety level	SIL 2

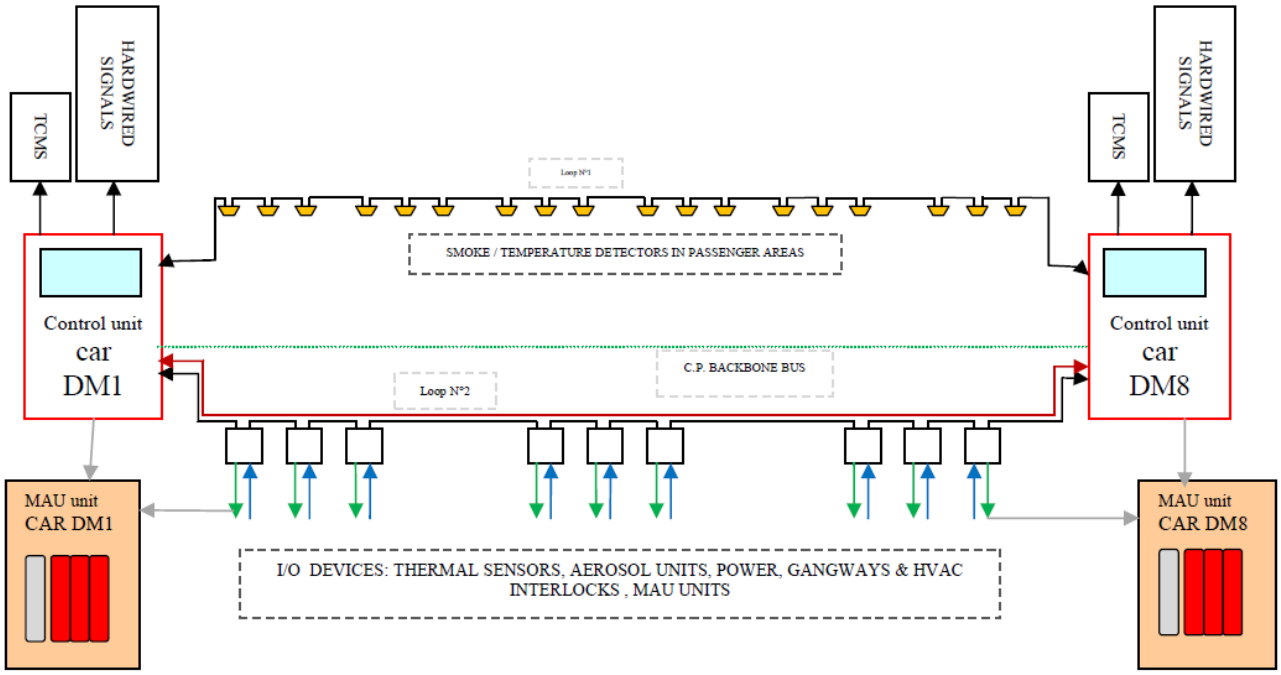
Electronic room is intended the area between the driver cab and the passenger saloon.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.71	One smoke/temperature detector shall be foreseen in each electronic room.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.139	These detectors shall be connected to the control units through the passenger areas detectors line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

3.2 Supervision of the fire detection system

3.2.1 Function Design

The fire protection system is controlled and commanded by two control units installed in car DM1 and car DM8 and working in Master-Slave configuration. The initial state of “master” or “slave” condition is set through a pre definite cabling on the interface connector that will set Master in car DM1 and slave in car DM8. Then the two control units will alternate periodically their statuses.
The two control units share two independent lines and exchange data on an additional independent serial line (backbone bus).
The first line connects all the detectors of the passenger areas while the second line connects a series of I/O modules. They have the function of collecting alarms from thermal detectors installed in technical areas, commanding consequently the aerosol cartridges and managing the car-based interlocks of HVAC, gangway doors and power disconnection.
In case of line interruption, both units will become Master in the sense that each control unit controls the devices respectively before and after the cable interruption.
In case of short circuit on the line, the interested piece of electric cable will be isolated by the isolators equipping every device on the line, and again both units will become Master, controlling all the devices respectively before and after the isolated cable in short circuit.
In both cases the control units will exchange all relevant data and will align through the independent and dedicated serial line so that the communication with TCMS is done only via the Master control unit.
The Picture 2 below schematically shows the configuration of the system.



Picture 2 -- Configuration of the system

3.2.1.1 Normal Condition

The two control units work in Master/Slave configuration: the Master control unit becomes slave if a fault occurs or after a pre definite period of time. The two control units receive the information about the smoke/temperature levels from the detectors in the passenger areas through the detectors line and receive the information/send the commands from/to the LHC and aerosol in the technical areas through the I/O modules line.

3.2.1.2 Failure Cases

a) Loss of function of a control unit.

b) Interruption/short circuit of a line.

3.2.1.3 Failure Detection

a) The presence of power supply is monitored as well as the functionality of the relays performing safety functions. Therefore, failures are transmitted to TCMS.

b) Failures are transmitted to TCMS.

3.2.1.4 Failure Handling

a) One redundancy is lost but the complete control of the system is automatically taken by the other control unit so that the complete system functionality is preserved.

b) The two control units become Master and no information is lost. Line cabling needs to be checked.

3.2.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.99
Description	All smoke/fire detectors shall be supervised by the FPS
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.72	Two control units shall be foreseen working in Master-Slave configuration.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.146	Control units will be identified by a vehicle cabling (jumper) on the interface connector.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.147	The Master control unit shall become Slave if a fault occurs or after a predefined operation time.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.465	The commutation between Master and Slave control units shall be managed in a safe way avoiding indefined configurations that can endanger the safety of the system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.332	Both control units shall monitor the presence of the power supply.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.334	In case the control unit realizes that no power supply is available, a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.333	TCMS shall store an event (Maintenance) in case of power supply is missing on either of the control units.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.148	DELETED	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.149	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.150	TCMS shall monitor the communication status of both Master and Slave control unit.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.151	If TCMS finds that there is no communication with either of the control units, an event (Maintenance) shall be stored.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.152	All smoke/temperature detectors (passenger saloon, vestibules, toilets, driver cabs and electronic rooms) shall be connected to both the control units through a communication line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.153	I/O modules shall be foreseen in order to manage the technical areas of the coach (read the status of thermal detectors, command aerosol cartridges) and the local commands to other systems (HVAC, gangway doors, power supply).	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.154	I/O modules shall be connected to both the control units through a communication line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.336	Each I/O module shall monitor the presence of the power supply.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.337	In case the I/O module realizes that no power supply is available, a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.338	TCMS shall store an event (Maintenance) in case of power supply is missing on an I/O module.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.155	The two control units shall share also a communication (RS485) line as a back up.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.45
Description	Failure in the communication lines shall be detected.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.73	The control units shall be able to detect a failure in the detectors line as well as in the I/O modules line.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.156	A single point failure in the line shall not affect the communication between either the detectors and the control units or the I/O modules and the control units.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.157	Failures in the line shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.158	TCMS shall store an event (Maintenance) in case of a failure in the lines.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.100
Description	A failure on a single detector shall not affect the rest of the fire detection system. A single point of failure of the communication line of fire detectors shall not cause the function to fail.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.74	A failure on a smoke/temperature detector shall be detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.159	A failure on a single detector in passenger areas shall not affect the rest of the fire detection system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.160	A failure on a LHC shall be detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.104
Description	The fire detection system shall operate in all train modes that have power supply available.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.75	The control units of the fire detection system shall be powered in all train modes that have power supply available.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.3 Driver interface

3.3.1 Smoke/heat alarm

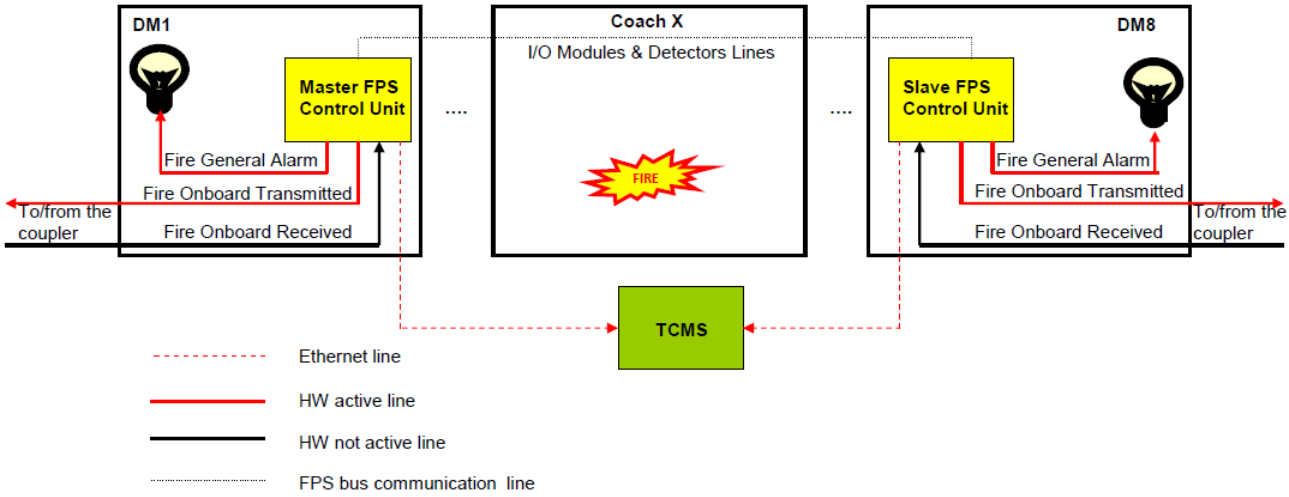
The fire detection system of the passenger compartment is able to recognize a Pre-alarm status, i.e. a suspicious status where a level of smoke/temperature higher than the nominal is detected. As soon as the level of smoke/temperature exceeds the alarm level the system will get to Alarm status.
The thermal detectors installed in the technical areas are only able to detect the Alarm status.

In case of fire Pre-alarm status, the system will inform TCMS about which coach is in pre-alarm. TCMS according the received information will command the HVAC unit of the coach where the pre-alarm was recognize to work in HVAC_Full_Recirculation_Mode.
In case of fire Alarm status is detected in the passenger areas or in any technical areas of a coach, following actions are performed:
1. TCMS is informed and a suitable lamp on all driver desks is lit;
2. both HVAC units of the coach where the fire has been detected are commanded to work in HVAC_Fire_Mode (i.e. ventilation is shut off and external shutters are closed); the HVAC of the other coaches are commanded to work in HVAC_Full_Recirculation_Mode (realized by TCMS);
3. both gangway doors of both gangway of the coach where the fire has been detected are commanded to close;
4. if the fire has been detected in the passenger areas, power is interrupted to the equipments positioned inside that coach which can cause electrical shock in case of water mist discharge or damages to components (realized by TCMS); if the fire has been detected in a technical area, power is shut down to the involved equipment in order not to feed the fire.

3.3.1.1 Function Design

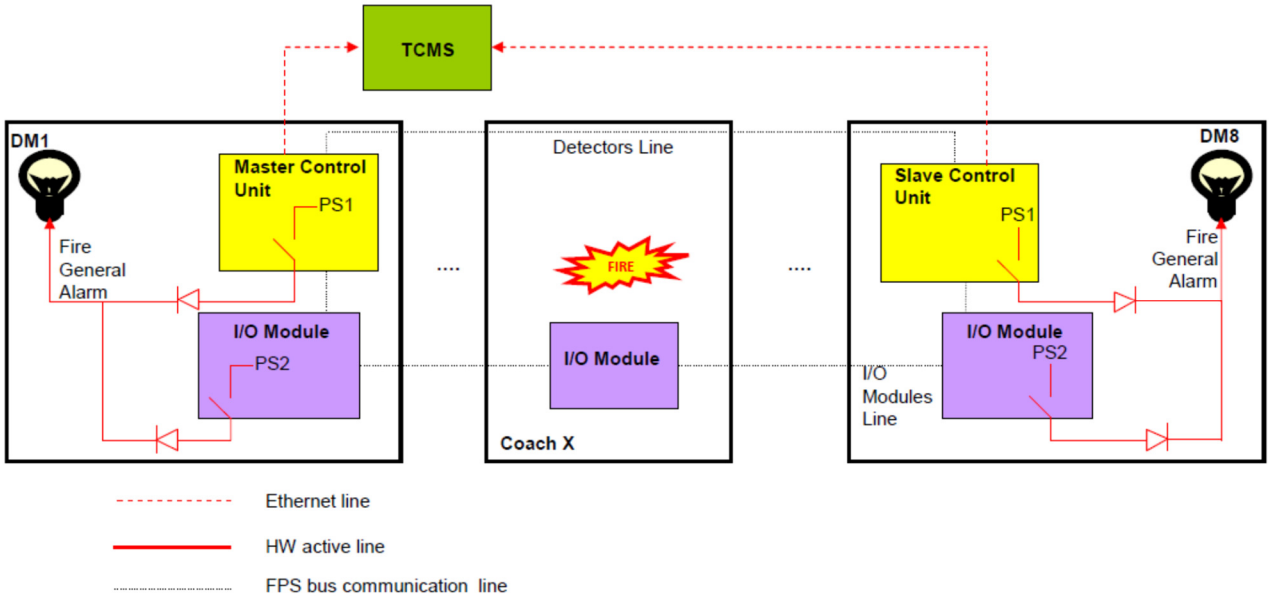
In order to perform the above described action 1. the Master control unit is alarmed via the information received by the detectors line (passenger areas) or the I/O modules line (technical areas) and activates the HW Fire General Alarm signal (the signal is generated redundantly by the Master control unit and by one I/O module). This signal will light a lamp on the driver desk of the coach where the control unit is mounted. Moreover the Master control unit will communicate the alarm via Ethernet to TCMS which generates an Event on the IDU. The Slave control unit will also activate the HW Fire General Alarm in order to light a lamp on the driver desk of the coach where it is mounted.

In order to light the lamp also on the driver desks of a possible coupled train, 2 HW signals are foreseen, the first, Fire Onboard Transmitted (generated redundantly), for sending the information that the own train is on fire and the second, Fire Onboard Received, for receiving the information that the other train is on fire. The 2 signals are sent over the coupler and the Fire Onboard Transmitted of one train is connected to the Fire Onboard Received of the other train.
Picture 3 below shows the involved vehicle interface signals in the function generation:



Picture 3 - Fire General Alarm signal generation

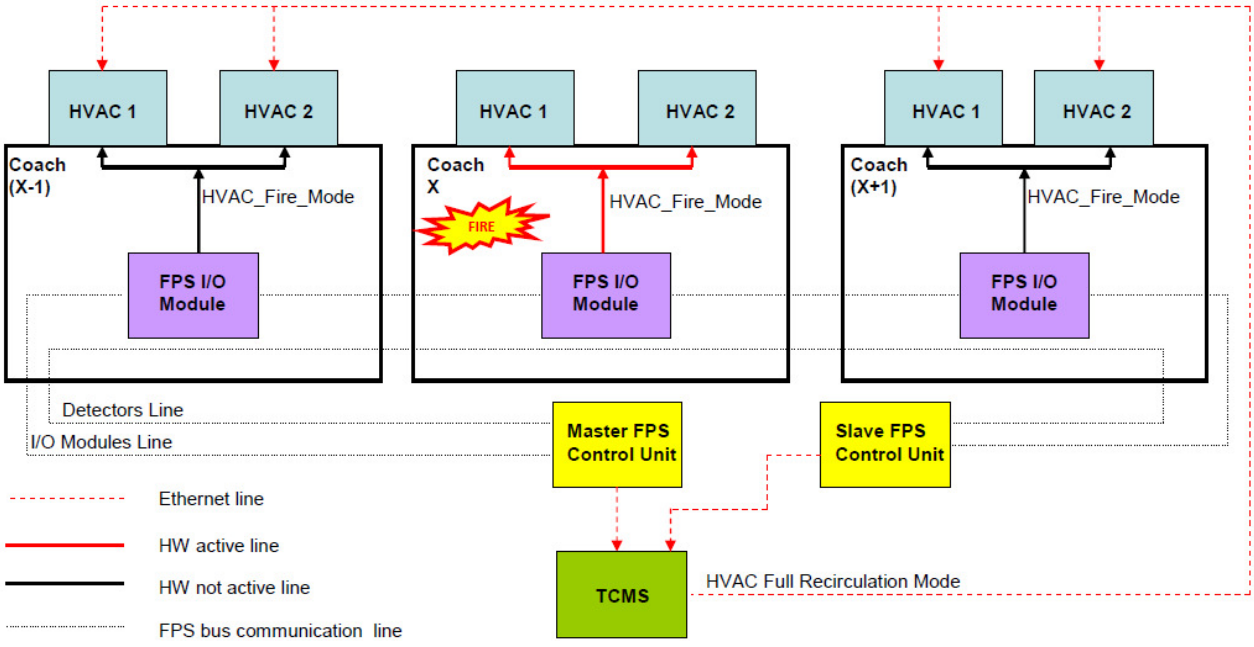
In particular, each Fire General Alarm signal is managed by a positive logic redundantly generated by HW circuits (NO contact of a supervised relay) in the control unit and in the relevant I/O module, both powered by different power supplies. In order to check the availability of the lamp power supply, a periodic check of the two feeding lines is foreseen.
Picture 4 below shows schematically the circuits involved in the positive activation of the lamp on the driver desks:



Picture 4 – Positive activation of the lamp

In order to perform the above described action 2., the Master control unit, as soon as alarmed by the information received by the detectors line (passenger areas) or the I/O modules line (technical areas), commands the relevant I/O modules to activate the HW signal for commanding the HVAC of the coach where fire has been detected to work in HVAC_Fire_Mode (the signal is generated redundantly by 2 different I/O modules). The HVAC_Full_Recirculation_Mode of the HVAC units of all the other coaches is set by TCMS upon the information received by the FPS.

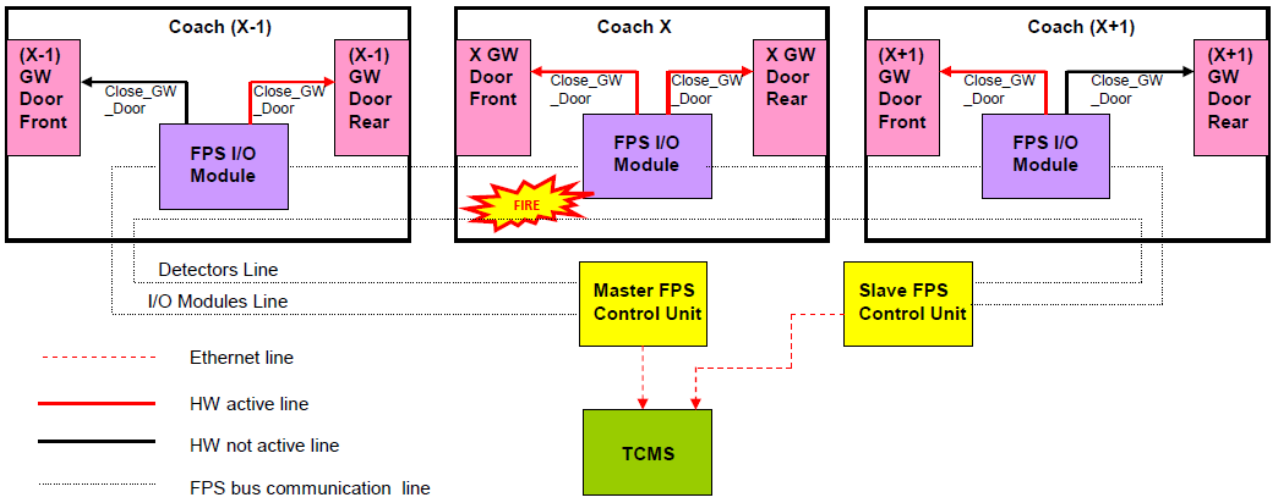
Picture 5 below shows the involved vehicle interface signals in the function generation:



Picture 5 - HVAC command signal generation

In order to perform the above described action 3., the Master control unit, as soon as alarmed by the information received by the detectors line (passenger areas) or the I/O modules line (technical areas), commands the relevant I/O modules to activate the HW outputs (each output signal is generated redundantly by 2 different I/O modules) for

commanding the closure of the controlled gangway door
Picture 6 below shows the involved vehicle interface signals in the function generation:



Picture 6 - Gangway doors command signals generation

3.3.1.1.1 Normal conditions

As soon as the Master control unit detects a fire, it commands the already described outputs on itself and on the relevant I/O modules in order to lit the lamp on the driver desks and perform the coach selective interlocks (gangway doors, HVAC, power shut down). All these actions are performed redundantly by the control unit and one I/O module (lamp activation) or by two I/O modules of the affected coach (coach selective interlocks).

3.3.1.1.2 Failure cases

a) Loss of function of a control unit.
b) Loss of communication of a relevant I/O module.
c) Loss of function of a relevant I/O module.

d) Loss of function of the lamp on the driver desk.

3.3.1.1.3 Failure detection

a) See section 3.2.1.3
b) See section 3.2.1.3
c) The relays realizing the lamp activation and the interlocks are monitored by the control units; a failure is transmitted to TCMS.
d) Failure can only be detected during a suitable test that will be performed periodically (frequency to be defined).

3.3.1.1.4 Failure handling

a) See section 3.2.1.4

b) See section 3.2.1.4
c) One redundancy is lost but the function is not endangered. The faulty relay needs to be replaced.
d) The function is lost and the driver shall act according to the operational manual.

3.3.1.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.85
Description	When smoke/heat is detected in passanger area and/or in technical area this shall be indicated on the IDU as an event.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.76	TCMS shall store an Event (Alarm) in case of fire detected in any passenger or technical areas indicating which detector and coach is in alarm.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.86
Description	When smoke/heat is detected this shall be indicated to the driver via an optical signal hard wire to the fire detection system. The indication shall be done irrespective of activated cab.
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.77	The system shall provide 2 hard wired signals (_Fire_General_Alarm) carrying the information that a fire has been detected in any passenger areas or technical areas.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.169	The signals _Fire_General_Alarm shall not be affected by the fault of one control unit.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.170	Both signals _Fire_General_Alarm shall be provided in positive logic, i.e. energized as long as fire is detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.171	A suitable fail safe circuit shall be provided to switch ON a lamp on the driver desk of all cabs as long as the signals _Fire_General_Alarm are energized.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.172	The indication lamp on the driver desks indicating the status of the _Fire_General_Alarm signal shall be possible to test using the Lamp test button.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.360	The two internal circuits to the Fire Protection system managing the _Fire_General_Alarm signal generation shall be possible to test separately upon a command received by TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.361	The system shall sent to TCMS the result of the _Fire_General_Alarm test.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.362	TCMS shall send a command to the Fire Protection system in order to activate the test of the -_Fire_General_Alarm circuits.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.363	TCMS shall store an Event (Warning) if the result of the _Fire_General_Alarm test received from the Master control unit is negative	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.173	The information of fire detected (Alarm status) shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.174	The information of which detector is in Alarm status and which coach it belongs to shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.365	TCMS upon the receipt of the fire Alarm status shall command the HVAC systems of all the not affected coaches to work in HVAC_Full_Recirculation_Mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.175	The system shall provide, on each control unit, 2 hard wired signals (1 input, Fire_Onboard_Received, and 1 output, Fire_Onboard_Transmitted) for	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

	receiving/transmitting the information that on the coupled train/on the train there is a fire.				
2F_06.04.Zefiro-Europe.CONCEPT.176	The Fire_Onboard_Transmitted output signal shall be provided in negative logic, i.e. energized as long as no fire is detected in any passenger areas or technical areas.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.177	The Fire_Onboard_Transmitted signal shall not be affected by the fault of one control unit.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.178	The system shall energize the Fire_General_Alarm signal as soon as the Fire_Onboard_Received is low but shall not activate the fire suppression/extinguishing system on the unaffected vehicle.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.179	The 2 hard wired signals for receiving/transmitting the information of Fire_Onboard_Received and Fire_Onboard_Transmitted from one train to the other shall be sent over the coupler.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.180	Suitable HW logics shall be foreseen to enable the transfer of the Fire_Onboard_Received and Fire_Onboard_Transmitted information when a second train is coupled.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.181	When a suspicious level of smoke/heat is detected in the passenger areas of a coach, the system shall transmit to TCMS the Pre-alarm status of the system.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.183	TCMS shall store an Event (Warning) in case the information of Pre-alarm status is received indicating which coach is in Pre-alarm.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.184	TCMS upon the receipt of the fire Pre-Alarm status of a coach shall command the HVAC system of that coach to work in HVAC_Full_Recirculation_Mode.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.87
Description	When smoke/heat is detected an acoustic alarm shall be started in the activated cab.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.78	TCMS shall set the buzzer in the active driver cab in case of fire detected in any passenger or technical areas.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.185	A buzzer shall be foreseen in order to realize the acoustical warning for the driver in case of fire.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	N/A

ID	2F_06.04.Zefiro-Europe.TRS.88
Description	When smoke/heat is detected the HVAC shall be commanded to “fire mode” via a hard wired signal
Safety level	SIL 1

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.79	One hardwired output signal for each coach (8 per vehicle) shall be provided to command the HVAC_Fire_mode of the coach where fire has been detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.186	Suitable HW logic shall be provided to transmit the command from the 8 outputs of the Fire Protection System to the corresponding HVAC interfaces.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.187	All the 8 hardwired signals shall be energized as long as no fire is detected and shall not be energized as soon as smoke/heat is detected in	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2

	the corresponding coach and kept in this staus until the system is reset.				
2F_06.04.Zefiro-Europe.CONCEPT.487	When smoke/heat is detected by a smoke detector located in vestibules or toilets, the 2 hardwired signals related to HVAC of the coach where the detector is installed and of the adjacent coach (on the side of the smoke detector) shall be not energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.188	Each HVAC unit shall work in HVAC_Fire_Mode (shut down, disconnect heaters and close external shutters) as soon as the hardwired signal from the Fire Protection System is not energized.	Derived Requirement	4S_13.-.HVAC		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.189	The status of the output command for each HVAC shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.190	TCMS shall acquire the status of each HVAC_Fire_Mode command provided by the FPS.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.191	The system shall monitor each HVAC_Fire_Mode output command comparing the status and the input of the relay used to implement the function. In case of incoherence the system shall transmit a message of fault to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.192	TCMS shall store an Event (Maintenance) in case of fault of the relay realizing the function of commanding the HVAC systems in HVAC_Fire_Mode.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.109
Description	When smoke/heat is detected the gangway doors shall be commanded to "fire mode" via a hard wired signal
Safety level	SIL 1

When smoke/heat is detected in a coach, 2 or 4 hardwired output signals will be activated by the Fire Protection System to command the closure of the corresponding relevant gangway doors.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.91	One hardwired output signal for each gangway door (14 per vehicle) has to be provided to command the closure of such door in case of fire in the relevant or adiacent coach.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.194	Suitable HW logic has to be foreseen to transmit the command from the 14 outputs to the corresponding gangway doors.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.195	When smoke/heat is detected in coach DM1 the 2 hardwired signals (DM1_ClsGwDoor_rear and TT2_ClsGwDoor_front) related to the gangway doors between DM1 and TT2 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.196	When smoke/heat is detected in coach TT2 the 4 hardwired signals (DM1_ClsGwDoor_rear, TT2_ClsGwDoor_front, TT2_ClsGwDoor_rear and M3_ClsGwDoor_front) related to the gangway doors between DM1 and TT2 and between TT2 and M3 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.197	When smoke/heat is detected in coach M3 the 4 hardwired signals (TT2_ClsGwDoor_rear, M3_ClsGwDoor_front, M3_ClsGwDoor_rear and T4_ClsGwDoor_front) related to the gangway doors between TT2 and M3 and between M3 and T4 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.198	When smoke/heat is detected in coach T4 the 4 hardwired signals (M3_ClsGwDoor_rear and T4_ClsGwDoor_front, T4_ClsGwDoor_rear and T5_ClsGwDoor_rear) related to the gangway doors between M3 and T4 and between T4 and T5 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1

2F_06.04.Zefiro-Europe.CONCEPT.199	When smoke/heat is detected in coach T5 the 4 hardwired signals (T4_ClsGwDoor_rear and T5_ClsGwDoor_rear, T5_ClsGwDoor_front and M6_ClsGwDoor_rear) related to the gangway doors between T4 and T5 and between T5 and M6 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.200	When smoke/heat is detected in coach M6 the 4 hardwired signals (T5_ClsGwDoor_front and M6_ClsGwDoor_rear, M6_ClsGwDoor_front and TT7_ClsGwDoor_rear) related to the gangway doors between T5 and M6 and between M6 and TT7 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.201	When smoke/heat is detected in coach TT7 the 4 hardwired signals (M6_ClsGwDoor_front and TT7_ClsGwDoor_rear, TT7_ClsGwDoor_front and DM8_ClsGwDoor_rear) related to the gangway doors between M6 and TT7 and between TT7 and M8 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.202	When smoke/heat is detected in coach DM8 the 2 hardwired signals (TT7_ClsGwDoor_front and DM8_ClsGwDoor_rear) related to the gangway doors between TT7 and DM8 shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.203	All the 14 hardwired signals shall not be energized as long as no smoke is detected, shall be energized as soon as smoke is detected and shall be kept energized until the system is reset.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.485	When smoke/heat is detected by a smoke detector located in vestibules or toilets, the hardwired signals related to gangway doors of the coach where the detector is installed and of the adjacent coach (on the side of the smoke detector) shall be energized.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.204	Each gangway door shall close as soon as the hardwired signal from the Fire Protection System is energized.	Derived Requirement	4S_08.06.02.-.Gangway Doors		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.205	The status of the output command for each gangway door shall be transmitted to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.206	TCMS shall receive the status of the command given by the Fire Protection System to each gangway door.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.207	The system shall monitor each output command comparing the status of the related relay and the input. In case of incoherence the system shall transmit a fault message to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.208	TCMS shall store an event (Maintenance) in case of fault of the relay realizing the function of closing gangway doors .	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.114
Description	When smoke/heat is detected in the passenger areas, power supply shall be interrupted to the equipment that can cause electrical shock.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.370	TCMS upon the information received from the Fire Protection System shall command the power supply shut off of at least following equipments positioned in the coach where fire is detected: · plugs on the seats (where present); · reading lights on the seats (where present); · PIS monitors; · lighting system (except for emergency functions); · MV electrical devices in the toilets;	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

	· MV electrical devices in the bistro area (if M3 is affected).				

ID	2F_06.04.Zefiro-Europe.TRS.110
Description	When smoke/heat is detected in the underfloor technical area power supply shall be interrupted to the technical area.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.120	One hardwired output signal for each coach (8 per vehicle) shall be provided to shut off the power supply to the technical area where fire has been detected.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.213	Suitable HW logic shall be provided to transmit the command from the 8 outputs of the Fire Protection System to the relevant contactors that interlock the power supply to the affected area while running capability shall be preserved.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_06.04.Zefiro-Europe.CONCEPT.214	All the 8 hardwired signals shall be provided as free voltage contacts managed in negative logic, i.e. energized as long as no fire is detected in any technical area of the coach.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.215	The system shall monitor each output signal comparing the status and the input of the relay used to implement the function. In case of incoherence the system shall transmit a message of fault to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.216	TCMS shall store an Event (Maintenance) in case of fault of the relay realizing the function of shutting off the power supply to the technical areas.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.369	TCMS upon the information received from the Fire Protection System shall command the power supply switch off to the affected area.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.2 Mute function

3.3.2.1 Function Design

Due to the fact that the fire alarm is classified as an Alarm Event, the mute function is automatically performed by the driver when he acknowledges the Event.

3.3.2.1.1 Normal conditions

In normal condition the fire warning buzzer can be muted by a suitable command available to the driver. This is not a safety relevant function and the causes of possible loss of the function are not under the fire protection system control.

3.3.2.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.73
Description	The driver shall have the possibility to mute sound alarm in case of smoke/heat detection.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL

2F_06.04.Zefiro-Europe.CONCEPT.80	The same button used to mute other acoustical warnings in the driver cab can be used to mute also the fire buzzer.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.3.3 Reset function

3.3.3.1 Function Design

A suitable push button is foreseen on each control unit in order to electrically reset the system. The electrical reset of the system involves the reset of the status of the smoke/temperature detectors for passenger areas (if any was activated), the reset of the Fire_General_Alarm and Fire_Onboard_Transmitted signals and of the commands to HVAC, gangway doors and power disconnection systems.

3.3.3.1.1 Normal conditions

As soon as the push button on one of the two control units is pressed and fire conditions are no longer present, the system will reset the Alarm status.

3.3.3.1.2 Failure cases

a) Fault on the push button.
b) Stuck contact of one of the relays generating the _Fire_General_Alarm or the gangway doors management functions.

3.3.3.1.3 Failure detection

a) Electrical continuity of the push button is monitored by an end of line resistor.
b) Relay implementing the function is monitored, TCMS is informed. The lamps on the driver desks are permanently lit.

3.3.3.1.4 Failure handling

a) The function is lost and the driver shall act according the operational manual.
b) The function is lost and the driver shall act according the operational manual.

3.3.3.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.41
Description	The smoke/heat optical signal shall be switched off when the system is reset.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.81	A push button on both control units shall be foreseen in order to reset the system following an alarm.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.433	Each push button shall be monitored. In case of fault a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.434	TCMS shall store an event (Maintenance) in case of fault of the reset push button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.219	As soon as the push button is pressed on either of the control units and fire conditions are no longer present, the electrical reset of the system shall be accomplished.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.220	Electrical reset of the system shall include: · reset of the status of smoke/temperature detectors of passenger areas; · reset of the Fire General Alarm signal; · reset of the Fire_Onboard_Transmitted signal; · reset of the HVAC_Fire_Mode signals; · reset of the commands towards gangway doors; · reset of commands for shutting down power supplies.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.84
Description	It shall not be possible to switch off smoke/heat alarm until the system is reset.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.82	Status of the _Fire_General_Alarm and Fire_Onboard_Transmitted signals shall be reset when the reset push button on either one of the the control unit is pressed and fire conditions are not present.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 2
2F_06.04.Zefiro-Europe.CONCEPT.435	The signal _Fire_General_Alarm shall not be de activated by any HW or SW circuits.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.3.4 Display detectors status on IDU

3.3.4.1 Function Design

This function provides all the information regarding the status of the complete system and the system detectors to the TCMS in order to be displayed on a suitable IDU page. This is not a safety relevant function.

3.3.4.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.64
Description	The status of the fire detection system shall be indicated on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.479	The system shall transmit to TCMS its functional status using the ICD available variable IGenSystemMode.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.83	The system shall transmit to TCMS its status (Starting, Stand-by, Pre-alarm, Alarm, Suppression, Extinguishing (after delay time), Discharge disabled, Maintenance/Test). *The statuses of Suppression and Discharge disabled are no applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.221	TCMS shall store the Fire Protection System status (Starting, Stand-by, Pre-alarm, Alarm, Suppression, Extinguishing (after delay time), Discharge disabled, Maintenance/Test) and show it on a suitable IDU page.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

	The statuses of Suppression and Discharge disabled are no applicable to ETR1000 Spain.				
2F_06.04.Zefiro-Europe.CONCEPT.480	TCMS shall store the functional status of the Fire Protection System using the variable IGenSystemMode received by the Master control unit which is identified by the variable ISpecSystOkMode=0.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.481	TCMS shall lit the "Out of order" dedicated icon on the IDU as soon as the Master control unit provides the information that the functional status of the system is set to Out of Order or Test/Maintenance (IGenSystemMode=3 OR 10) or when both main circuit breakers relevant to the control units are tripped	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.482	TCMS shall store an Event (Warning) in case the functional status of the system is set to Out of Order (IGenSystemMode=3).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.483	TCMS shall store an Event (Warning) in case the functional status of the system is set to Test/Maintenance (IGenSystemMode=10).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.484	TCMS shall store an Event (Warning) in case the functional status of the system is set to Degraded (IGenSystemMode=2).	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.68
Description	The status of the fire/smoke detectors shall be indicated on the IDU (detecting smoke/heat, detector faulty).
Safety level	SIL 0

The LHC and the aersol cartridges need to be replaced after an alarm.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.84	The system shall transmit to TCMS the status of all detectors installed in the vehicle: · smoke/temperature detectors (Normal, Smoke alarm, Thermal alarm, Fault, Sensor disabled); · linear heat cables (Normal, Alarm, Fault).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.223	TCMS shall receive the status of all detectors installed in the vehicle: · smoke/temperature detectors (Normal, Smoke alarm, Thermal alarm, Fault, Sensor disabled); · linear heat cables (Normal, Alarm, Fault) and show it on a vehicle synoptic chart on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.224	TCMS shall store an Event (Maintenance) in case the status of a smoke/temperature detector of the passenger areas is set to Disabled or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.225	TCMS shall store an Event (Maintenance) in case the status of a LHC is set to Alarm or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.466	The system shall transmit to TCMS the status of all aerosol cartridges installed in the vehicle (Normal, Activated, Fault).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.467	TCMS shall receive the status of all aerosol cartridges installed in the vehicle (Normal, Activated, Fault) and show it on a vehicle synoptic chart on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.468	TCMS shall store an Event (Maintenance) in case the status of an aerosol cartridge is set to Activated or Fault.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

3.3.5 Interrupt fire extinguisher

This functionality is not applicable to ETR1000 Spain.

3.3.5.1 Function Design

3.3.5.2 Normal condition

As soon as the push button is pressed on either one of the control units, all the electrovalves on the MAU units will close and the water mist discharge is stopped.

3.3.5.3 Failure cases

a) Electrical continuity of the push button not found.
b) Stuck contact of the push button.

3.3.5.4 Failure detection

a) Push button is monitored by an end on line resistor. Failure is transmitted to TCMS.
b) A permanent indication of pushed button (water mist interruption requested) is transmitted to TCMS as a fault.

3.3.5.5 Failure handling

a) Function is endangered but loss of safety is not involved. Attended during maintenance.
b) In case a permanent indication of interruption is present, the suppression function is endangered and the driver shall act according the operational manual.

3.3.5.6 Requirements

All the requirements included in this section are not applicable to ETR1000 Spain.

ID	2F_06.04.Zefiro-Europe.TRS.103
Description	The driver shall be able to interrupt the fire extinguishing in case of a false alarm.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.85	A suitable push button shall be foreseen on the control unit for interrupting the water mist disharge.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.474	Electrical continuity of the push button shall be monitored. In case of fault a diagnostic message shall be sent to TCMS.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.473	TCMS shall store an event (Maintenance) in case of fault of the abort push button.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.472	The system shall send a diagnostic message in case the push button turns out to be pressed more than a reasonable time.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

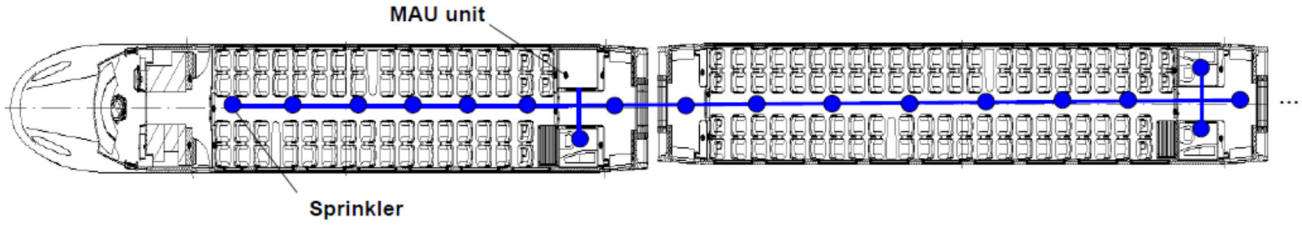
2F_06.04.Zefiro-Europe.CONCEPT.475	DELETED	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.226	The command for interrupting the water mist discharge shall be managed with positive logic.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

3.4 Fire Extinguisher

Fire suppression system in passenger areas is not applicable to ETR1000 Spain.

3.4.1 Function Design

The fire suppression system in passenger areas is based on high pressure water mist technology. Main components of the system are: MAU units (containing water cylinders and pressurized gas cylinders) and the stainless steel piping network with suitable nozzles for the water mist discharge.
Two MAU units are foreseen, one positioned in DM1 and one in DM8. The piping network is constituted by a main branch connecting the 2 MAU units along the whole train and a secondary pipe in each coach where special nozzles, of sprinkler type including a thermal sensitive bulb, are connected.
In the Picture 7 below the position of the sprinklers and the MAU unit is shown in the first 2 coaches; in the other coaches the devices will be positioned symmetrically except for DM3 which will present a special configuration of sprinklers due to the particular layout of the coach.



Picture 7 - Sprinklers and MAU unit layout

The fire extinguishing system in the technical areas is performed through aerosol cartridges of suitable dimensions installed inside each box and activated selectively (Extinguishing status) by the control unit following the information coming from the detection system and using the I/O modules for the communication.

3.4.1.1 Normal condition

In stand-by conditions, the piping in the passenger areas is dry and the nozzles are hidden in the ceiling.
As soon as the Master control unit detects a fire (fire Alarm status), the valves on both the cylinder units are commanded to open and the water now pressurized will flow in the main and secondary pipes. As a consequence the thermal bulbs of all the vehicle sprinklers will pop out mechanically and will get visible (Pre-activation status). Anyway the water mist discharge will happen only through those sprinklers whose heat sensitive bulb has broken due to the high temperature (Suppression status).

As far as the technical areas extinguishing system is concerned, as soon as the Master control unit detects a fire in a technical area, the aerosol of that area is commanded to open.

3.4.1.2 Failure cases

a) Loss of function of a MAU electrovalve.
b) Breaking of a sprinkler.
c) Loss of piping integrity.
d) Low pressure on the gas cylinder.
e) Low level on the water cylinders.

f) Loss of function of a aerosol activator.

3.4.1.3 Failure detection

a) Electrical continuity of each electrovalve is monitored by the control units and therefore the TCMS is informed.
b) No diagnostic is implemented. Periodic checks are foreseen.
c) No diagnostic is implemented. Periodic checks are foreseen.
d) Pressure level in the gas cylinders is monitored by the control units and therefore the TCMS is informed.
e) Only a visual level indicator is present, periodic checks are foreseen.
f) Each aerosol activator is monitored by the control units and therefore the TCMS is informed.

3.4.1.4 Failure handling

a) A redundancy is lost but the function is not endangered. The driver shall act according the operational manual.
b) Function is endangered. Attended during maintenance as soon as discovered.
c) Function is endangered. Attended during maintenance as soon as discovered.
d) The function is executed by the other MAU unit but not correctly. The driver shall act according the operational manual.
e) Function is executed by the other MAU unit but not correctly. Attended during maintenance as soon as discovered.
f) The function is lost and the driver shall act according the operational manual.

3.4.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.93
Description	When smoke/heat is detected the fire extinguisher for that specific area shall be activated.
Safety level	SIL 1

Following electrovalves opening, the system is in pre-activated status, i.e. pressurized water flows in all pipes and all the sprinklers pop out from the ceiling.

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.86	As soon as the fire detection system detects a fire in a technical area, the aerosol cartridge installed in that area shall be commanded to open after a suitable delay time (TBD).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.230	The information that an aerosol cartridge has been commanded to discharge shall be transmitted to TCMS (Extinguishing status).	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.231	The information of which aerosol cartridge has been commanded to discharge shall be transmitted to TCMS by setting the status of that cartridge to Activated.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.233	As soon as the fire detection system detects a fire in passenger areas, both electrovalves on both the MAU units shall be commanded to open. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 1
2F_06.04.Zefiro-Europe.CONCEPT.234	DELETED	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

2F_06.04.Zefiro-Europe.CONCEPT.236	The system shall transmit to TCMS the status of Suppression as soon as MAU electrovalves are open and at least one nozzle is open. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.94
Description	The fire extinguisher shall not be activated for areas were no smoke/heat is detected.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.87	Aerosol generators installed in technical areas where no fire is detected shall not be activated.	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A
2F_06.04.Zefiro-Europe.CONCEPT.237	Sprinklers shall be provided with a thermal sensitive bulb that breaks as the temperature reaches a certain threshold. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

ID	2F_06.04.Zefiro-Europe.TRS.95
Description	Fire extinguishing system shall perform a continuous self monitoring in order to detect possible failures such as: loss of gas/water in cylinders, incorrect position of valves, etc..
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.88	Electrical continuity of both gas actuation valves on the MAU units shall be monitored. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.238	In case no electrical continuity is found, a fault shall be transmitted to TCMS. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.239	TCMS shall store an Event (Maintenance) in case of fault of a gas actuation valve. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.240	The system shall monitor the pressure level in the gas cylinder of the MAU units. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.241	In case the pressure level goes below the functional value, a fault shall be transmitted to TCMS. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.242	TCMS shall store an Event (Maintenance) in case of loss of pressure of a gas cylinder. *This requirement is not applicable to ETR1000 Spain.*	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.243	Electric actuator of each aerosol cartridges shall be monitored.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.244	In case electrical continuity of the actuator is not found, the status of the corresponding cartridge shall be set to Fault .	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0

3.5 Maintenance/Test Mode of fire detection system

3.5.1 Function Design

3.5.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.81
Description	When the fire detection system is running in Maintenance/Test mode the TCMS shall not show any extra faults or alarms from the fire detection system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.89	The system shall communicate to TCMS that it is running on Maintenance/Test mode.	Derived Requirement	4S_06.05.-.Fire Protection Systems		SIL 0
2F_06.04.Zefiro-Europe.CONCEPT.246	TCMS shall not show any extra faults or alarms from the FPS as long as FPS status is Maintenance/Test.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_06.04.Zefiro-Europe.TRS.108
Description	It shall be possible to test the fire detection system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.93	It shall be possibile to test the FPS critic components (valves, detectors, pipes integrity).	Derived Requirement	4S_06.05.-.Fire Protection Systems		N/A

3.6 Events

3.6.1 Function Design

3.6.2 Requirements

ID	2F_06.04.Zefiro-Europe.TRS.113
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_06.04.Zefiro-Europe.CONCEPT.116	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.29	_Fire_General_Alarm_DM1	L	2F_06.04.Zefiro-Europe.CONCEPT.171
2F_06.04.Zefiro-Europe.CONCEPT.469	_Fire_General_Alarm_DM8	L	2F_06.04.Zefiro-Europe.CONCEPT.171
2F_06.04.Zefiro-Europe.CONCEPT.247	_Fire_Onboard_Transmitted_DM1	L	2F_06.04.Zefiro-Europe.CONCEPT.175
2F_06.04.Zefiro-Europe.CONCEPT.470	_Fire_Onboard_Transmitted_DM8	L	2F_06.04.Zefiro-Europe.CONCEPT.175
2F_06.04.Zefiro-Europe.CONCEPT.248	_Fire_Onboard_Received_DM1	L	2F_06.04.Zefiro-Europe.CONCEPT.175
2F_06.04.Zefiro-Europe.CONCEPT.476	_Fire_Onboard_Received_DM8	L	2F_06.04.Zefiro-Europe.CONCEPT.175
2F_06.04.Zefiro-Europe.CONCEPT.249	_DM1_HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.250	_TT2_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.251	_M3_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.252	_T4_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.253	_T5_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.254	_M6_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.255	_TT7_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.256	_DM8_ HVAC_Fire_Mode	L	2F_06.04.Zefiro-Europe.CONCEPT.79
2F_06.04.Zefiro-Europe.CONCEPT.257	_DM1_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.258	_TT2_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.259	_TT2_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.260	_M3_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.261	_M3_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194

2F_06.04.Zefiro-Europe.CONCEPT.262	_T4_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.263	_T4_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.264	_T5_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.265	_T5_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.266	_M6_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.267	_M6_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.268	_TT7_ClsGwDoor_front	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.269	_TT7_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.270	_DM8_ClsGwDoor_rear	L	2F_06.04.Zefiro-Europe.CONCEPT.194
2F_06.04.Zefiro-Europe.CONCEPT.271	_DM1_PS_ShutDown_Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.272	_TT2_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.273	_M3_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.274	_T4_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.275	_T5_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.276	_M6_PS_ShutDown_Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.277	_TT7_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.278	_DM8_PS_ShutDown_ Saloon	L	2F_06.04.Zefiro-Europe.CONCEPT.370
2F_06.04.Zefiro-Europe.CONCEPT.279	_DM1_PS_ShutDown_TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.280	_TT2_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.281	_M3_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.282	_T4_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.283	_T5_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120

2F_06.04.Zefiro-Europe.CONCEPT.284	_M6_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.285	_TT7_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120
2F_06.04.Zefiro-Europe.CONCEPT.286	_DM8_PS_ShutDown_ TechnicalAreas	L	2F_06.04.Zefiro-Europe.CONCEPT.120

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.30			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.31			

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.32	_Detectors_Line_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.158
2F_06.04.Zefiro-Europe.CONCEPT.287	_I/OModules_Line_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.158
2F_06.04.Zefiro-Europe.CONCEPT.288	_IP_Communication_Missed_Slave	L	2F_06.04.Zefiro-Europe.CONCEPT.151
2F_06.04.Zefiro-Europe.CONCEPT.289	_IP_Communication_Missed_Master	L	2F_06.04.Zefiro-Europe.CONCEPT.151
2F_06.04.Zefiro-Europe.CONCEPT.290	_SmokeDetector_XX_Status	A	2F_06.04.Zefiro-Europe.CONCEPT.76 2F_06.04.Zefiro-Europe.CONCEPT.174 2F_06.04.Zefiro-Europe.CONCEPT.223 2F_06.04.Zefiro-Europe.CONCEPT.224
2F_06.04.Zefiro-Europe.CONCEPT.291	_LHC_XX_Status	A	2F_06.04.Zefiro-Europe.CONCEPT.76 2F_06.04.Zefiro-Europe.CONCEPT.223 2F_06.04.Zefiro-Europe.CONCEPT.225
2F_06.04.Zefiro-Europe.CONCEPT.292	_Aerosol_Cartridge_XX_Status	A	
2F_06.04.Zefiro-Europe.CONCEPT.293	_Fire_General_Alarm	L	2F_06.04.Zefiro-Europe.CONCEPT.76
2F_06.04.Zefiro-Europe.CONCEPT.294	_Coach_in_Alarm	A	2F_06.04.Zefiro-Europe.CONCEPT.76 2F_06.04.Zefiro-Europe.CONCEPT.174
2F_06.04.Zefiro-Europe.CONCEPT.295	_Coach_in_Pre-alarm	A	
2F_06.04.Zefiro-Europe.CONCEPT.296	_System_Status	A	2F_06.04.Zefiro-Europe.CONCEPT.221

2F_06.04.Zefiro-Europe.CONCEPT.297	_HVAC_Relay_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.192
2F_06.04.Zefiro-Europe.CONCEPT.298	_GWDoor_Relay_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.208
2F_06.04.Zefiro-Europe.CONCEPT.300	_PS_ShutDown_TechnicalAreas_Relay_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.215
2F_06.04.Zefiro-Europe.CONCEPT.301	_MAU_Electrovalve_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.239
2F_06.04.Zefiro-Europe.CONCEPT.302	_Gas_Cylinder_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.242
2F_06.04.Zefiro-Europe.CONCEPT.456	_SCU_XX_PS_Missing	L	2F_06.04.Zefiro-Europe.CONCEPT.333
2F_06.04.Zefiro-Europe.CONCEPT.457	_IO_Module_XX_PS_Missing	L	2F_06.04.Zefiro-Europe.CONCEPT.338
2F_06.04.Zefiro-Europe.CONCEPT.461	_Fire_General_Alarm_Test_Result	L	2F_06.04.Zefiro-Europe.CONCEPT.363
2F_06.04.Zefiro-Europe.CONCEPT.458	_Reset_PB_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.434
2F_06.04.Zefiro-Europe.CONCEPT.459	_Abort_PB_XX_Continuously_Pressed	L	2F_06.04.Zefiro-Europe.CONCEPT.475
2F_06.04.Zefiro-Europe.CONCEPT.460	_Abort_PB_XX_Fault	L	2F_06.04.Zefiro-Europe.CONCEPT.473
2F_06.04.Zefiro-Europe.CONCEPT.478	_GWayDoor_XX_CmdState	L	2F_06.04.Zefiro-Europe.CONCEPT.206
2F_06.04.Zefiro-Europe.CONCEPT.477	_HVAC_XX_CmdState	L	2F_06.04.Zefiro-Europe.CONCEPT.190

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_06.04.Zefiro-Europe.CONCEPT.462	_Start_Fire_General_Alarm_test	L	2F_06.04.Zefiro-Europe.CONCEPT.362

5 References

[1] EC01P006793B - TRD Fire Detection and Extinguishing System